Sofija Dimitrijevic

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Education

California Polytechnic State University, San Luis Obispo Sep. 2022 – Jun. 2027
B.S./M.S. (Blended) in Computer Science with AI/ML Concentration, Minor in Linguistics GPA: 3.7

- Dean's List & President's Honor List (Fall 2022–Fall 2024, Spring 2025); SWE Scholarship Recipient 2025
- **Relevant Coursework:** Machine Learning, Applied Regression Analysis, Operating Systems, Software Engineering, Database Systems, Distributed & Parallel Computing, Computer Networks, Data Structures & Algorithms, Programming Languages, Systems Programming, Object-Oriented Programming + Design, Computer Security

Experience

Software Engineer Intern Jun. 2025 – Sep. 2025
HP Inc. *Houston, TX*

- Developed an **AI-powered Windows telemetry service** using a multi-class Random Forest model to predict PC power consumption based on user behavior, **improving AI model accuracy by 50%**. Applied extensive data visualization methods.
- Built a configurable **Windows desktop application** in C# and XAML, enabling **real-time** model inference and IPC messaging with backend service. Wrote custom benchmark testing for model predictions, with **90% accuracy**.

Physics Student Researcher Jul. 2024 – Sep. 2024
Cal Poly Partners *San Luis Obispo, CA*

- Integrated flocking into epithelial tissue dynamics with the Active Finite Voronoi (AFV) model to gain insights into cancer progression, **improved simulation efficiency by 10%** through advanced vectorization.
- Ported MATLAB model to Python using SciPy and Voronoi libraries for better maintainability and scalability.

Projects

MCP Server–Client Protocol | *Python, MCP, AsyncIO, Pydantic* Aug. 2025

- Built an asynchronous **Model Context Protocol** server and client using **Anthropic's MCP framework**, supporting custom tools, prompts, and document resources.
- Enabled concurrent tool calls and resource reads, **boosting throughput and responsiveness by 40%** compared to synchronous handling.

Gather: Digital Inventory Web App | *React, TypeScript, REST API* Apr. 2024 – Jun. 2024

- Developed a full-stack inventory management app with **CRUD** operations and **REST API** in an Agile team of 5.
- Built **full-stack** features and refactored backend into a reusable utility library, **cutting code duplication by 20%** and improving maintainability.

Moment: Measuring Burnout Web App (Hackathon) | *TensorFlow, scikit-learn* Jan. 2024

- Developed an AI-powered app to predict hospital staff burnout; implemented ML models for burnout scoring.
- Won **"Best Design"** at a 24-hour university-wide hackathon at Cal Poly.

Leadership & Campus Involvement

Women Involved in Software and Hardware (WISH) Sep. 2022 – Present
Vice President (Current), Technical Director (Previous) *San Luis Obispo, CA*

- Overseeing strategy and operations for a 300+ member organization; managing a 20-person officer board.
- Led a team of 4 to deliver 20+ events annually; organized programming workshops for members and newcomers.

Society of Women Engineers (SWE) Sep. 2022 – Present
Member; Scholarship Recipient *San Luis Obispo, CA*

Technical Skills

Languages: Python, Java, C/C++, C#, C, SQL, MATLAB, JavaScript, TypeScript, HTML, CSS, Racket, R
Libraries/Tools: Natural Language Toolkit, Hugging Face Transformers, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib, Optuna, CUDA, SciPy, Spotipy, Jupyter Notebook, .NET Framework, WPF, MCP, AsyncIO, Pydantic, Node.js, Express, MongoDB, Google Colab, Figma